Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	Yes, 10 antibodies detected, 0 RRID provided : anti-CD90-phycoerythrin : anti-CD44-allophycocyanin : anti-CD106-phycoerythrin-cyanine7 : PE-Cy7 : anti-CD45-fluorescein : anti-CD34-PE : anti-CD31-APC : anti-CD29-PE-Cy7 : anti-CD105-PE-Cy7 : anti-CD11b-FITC Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	Yes, 2 cell lines detected, 2 RRIDs provided Suggestion: RRID:CVCL_0063 : HEK293T Suggestion: RRID:CVCL_0030 : HeLa Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	Yes, 4 organisms detected, 1 RRID provided Suggestion: RRID:IMSR_JAX : C57BL6J : C57BL6J-Tg (SOD1G93A)1GurJ : C57BL6J-OlaHsd : NTg Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Reprogramming of human fibroblasts into neural progenitor cells (iNPCs) Skin biopsy collection from ALS patients and healthy donors (Table S1) was approved by the Research Ethics Committee of Sheffield (study number STH16573,	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
	reference 12YH0330) or by the Institutional Review Board of the Ohio State University (ethics number IRB08-00402) after informed consent was obtained.	
Provide statement confirming informed consent obtained from study participants.	Reprogramming of human fibroblasts into neural progenitor cells (iNPCs) Skin biopsy collection from ALS patients and healthy donors (Table S1) was approved by the Research Ethics Committee of Sheffield (study number STH16573, reference 12YH0330) or by the Institutional Review Board of the Ohio State University (ethics number IRB08-00402) after informed consent was obtained.	
Report on age and sex for all study participants.	**Age:**Materials and methods Isolation and expansion of mouse adipose-derived stem cells mADSCs were isolated from 8- to 10-week-old female C57BL6J mice as previously described with minor modifications. **Sex:**Materials and methods Isolation and expansion of mouse adipose-derived stem cells mADSCs were isolated from 8- to 10-week-old female C57BL6J mice as previously described with minor modifications.	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	Not detected.	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	For the in vivo therapeutic study, we followed the recommendations for the conduct of pre-clinical studies in the SOD1G93A mouse model released by the ALS Therapy Development Institute (ALS TDI), which were successively updated in 2009 following an international conference held in Germany, 11-13 June 2009. 74 To determine group size, a statistical power analysis using G*Power 3.1.	
Randomization	not detected.	
Blinding	All behavioral tests were performed by the same investigator, who was blinded to the treatment group.	
inclusion/exclusion criteria	not detected.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Passage 4 mAstrocytes from symptomatic (90 day old) SOD1G93A or NTg age-matched control mice were seeded in triplicate (technical replicate) onto 1 well of a 24-well plate previously coated with 1:1,000 poly-l-ornithine (Sigma, St. Louis, MO, USA) and laminin 1:200 (Sigma, St. Louis, MO, USA), at a concentration of 50,000 cellswell. mAstrocytes were maintained for 24 h in astrocyte basal medium (DMEM, 10 FBS, 1 PenStrep).	
Define whether data describe technical or biological replicates	Passage 4 mAstrocytes from symptomatic (90 day old) SOD1G93A or NTg age-matched control mice were seeded in triplicate (technical replicate) onto 1 well of a 24-well plate previously coated with 1:1,000 poly-l-ornithine (Sigma, St. Louis, MO, USA) and laminin 1:200 (Sigma, St. Louis, MO, USA), at a concentration of 50,000 cellswell. mAstrocytes were maintained for 24 h in astrocyte basal medium (DMEM, 10 FBS, 1 PenStrep).	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Reprogramming of human fibroblasts into neural progenitor cells (iNPCs) Skin biopsy collection from ALS patients and healthy donors (Table S1) was approved by the Research Ethics Committee of Sheffield (study number STH16573, reference 12YH0330) or by the Institutional Review Board of the Ohio State University (ethics number IRB08-00402) after informed consent was obtained.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Animal experiments All experiments involving mice were conducted in accordance with the Animals (Scientific Procedures) Act 1986 under a UK Home Office project license approved by the Sheffield University Ethical Review Committee Project Applications and Amendments Sub-Committee and by the UK Animal Procedures Committee (London, UK).	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	Not detected.	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	not detected.	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	A power of 80 was used with a two-tailed Students t test (a = 0.05, b = 0.8).	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	Not detected.	
If data are publicly available, provide accession number in repository or DOI or URL.	Not detected.	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	